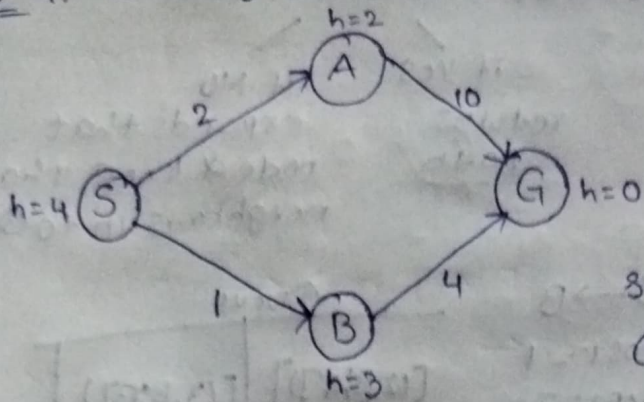


Lab-4: Greedy Best-First Search


$$S \rightarrow A \neq 2$$
$$S \rightarrow B = 1$$
$$A \rightarrow G = 10$$
$$B \rightarrow A = 4$$
$$\text{start} = 85$$

Goal = G .

Possible paths,

Path-1: $S \rightarrow A \rightarrow G$

Path-2: $S \rightarrow B \rightarrow G$

$$\text{cost} = 2 + 10 = 12$$
$$\text{cost} = 1 + 4 = 5$$

$5 < 12 \rightarrow$ best answer ($S \rightarrow B \rightarrow G$)

Heuristic: tells my guess of how close I'm to the goal.

Given, $h(S) = 4$ | $h(B) = 3$

$$h(A) = 2 \quad | \quad h(G) = 0$$

Smaller heuristic = looks closer.

$$\text{graph} = \{ 's' : [('A', 2), ('B', 1)] ,$$

$A': [C'G', 101]$

$$B' : [(G', u)],$$
 $G': []$

} Neighbours & cost

$$h = \{ 'S': 4, 'A': 2, 'B': 3, 'G': 0 \} \rightarrow \text{heuristic values.}$$

- We use queue so that we can traverse nodes level wise and can select best path on that level.

$$\text{queue} = [\text{ch}[\text{start}], [\text{start}]]$$

~~[cur, 'S', 'I']~~

Queue

$$Z = [C_u, [S']] \rightarrow h(S) = 4$$

- we continue these things until queue is empty.

queue.sort() → so that we can take the

Smallest one from that Queue.

∴ Parth = queue.pop(0) → takes 'S'

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currNode = S.
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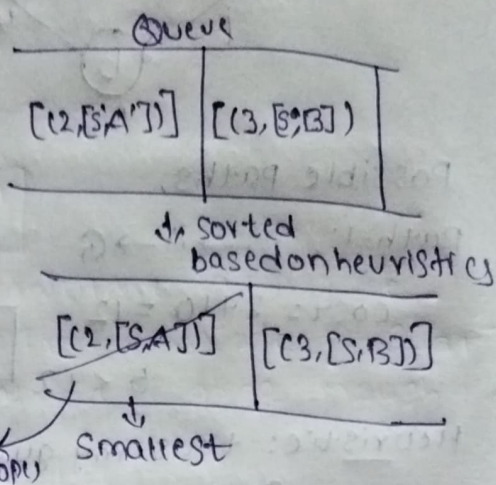
Checks, whether that $\text{currNode} == \text{goal}$

if Yes
returns path

if No
expands that node & keep that neighbours in to Queue.

$S \rightarrow A$
cost = 2
 $h(A) = 2$

$S \rightarrow B$
cost = 1
 $h(B) = 3$



path = $[S, A]$

node = $A \neq \text{goal}(G)$

↓
expand A.

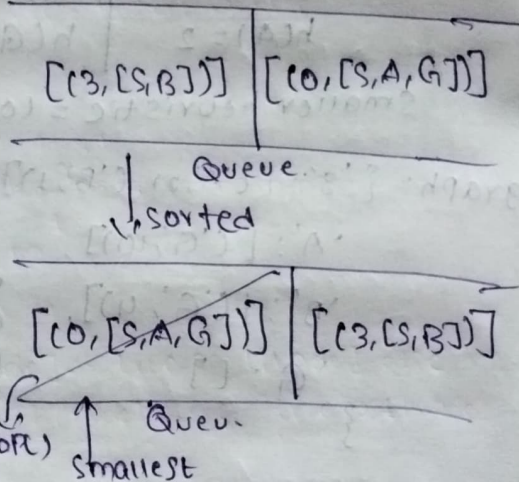
$A \rightarrow G = 10$

$h(G) = 0$

path = $[S, A, G]$

node = $G = \text{goal}(G)$

∴ Goal reached



Cost = $S \rightarrow A = 2$
 $A \rightarrow G = 10$ } $2 + 10 = 12$

path = $[S, A, G]$

Cost = 12

- But for path $[S, B, G]$

Cost = 5.

But, it is taking based on the heuristic values and for each node, it is taking a Greedy pick.

The line responsible for this is,

`queue.append((h[n], path+[n]))`

Why A^* does not make this mistake?

A^* uses $f(n) = g(n) + h(n)$

$g(n)$ = cost already traveled

$h(n)$ = estimated cost remaining

- Greedy uses only heuristic values and ignores the cost

$$f(n) = g(n) + h(n)$$